JAVA PROGRAM SUMS(11/7/26)

1. TO FIND AVERAGE:

Main.java

```java
class Main
{
public static void main(String[] args)
{
      int TAMIL = 50;
      int ENGLISH = 48;
      int MATHS = 45;
      int SCIENCE = 42;
      int SOCIAL = 50;
      int average = (TAMIL+ENGLISH+MATHS+SCIENCE+SOCIAL)/5;
      int total = TAMIL+ENGLISH+MATHS+SCIENCE+SOCIAL;
      System.out.println("AVERAGE MARKS: "+average);
      System.out.println("total marks out of 250: "+total);

}
}
```

```
AVERAGE MARKS: 47
total marks out of 250: 235
```

2.TO FIND SIMPLE INTEREST:

Run Debug Stop Share Save { } Beau

Main.java

```java
class Main
{
public static void main(String[] args)
{
      int p = 3;
      int r = 3;
      int t = 3;
      int si = (p*r*t)/100;
      System.out.println("simple interest: "+si);

}
}
```

```
simple interest: 0
```

3. TO FIND STUDENT'S RESULT:

Main.java

```java
class Main
{
public static void main(String[] args)
{
      int TAMIL = 50;
      int ENGLISH = 48;
      int MATHS = 45;
      int SCIENCE = 42;
      int SOCIAL = 50;
      int average = (TAMIL+ENGLISH+MATHS+SCIENCE+SOCIAL)/5;
      int total = TAMIL+ENGLISH+MATHS+SCIENCE+SOCIAL;
      System.out.println("AVERAGE MARKS: "+average);
      System.out.println("total marks out of 250: "+total);

}
}
```

```
AVERAGE MARKS: 47
total marks out of 250: 235
```

4. ARITHMETIC OPERATORS:

Main.java

```java
class Main
{
public static void main(String[] args)
{
      int a = 3;
      int b = 3;
      int sum = a+b;
      int subtraction = a-b;
      int multiplication = a*b;
      int division = a/b;
      int modulus = a%b;
      System.out.println("sum:"+ sum);
      System.out.println("subtraction:"+ subtraction);
      System.out.println("multiplication:"+multiplication);
      System.out.println("division:"+division);
      System.out.println("Modulus:"+ modulus);

}
}
```

```
subtraction:0
multiplication:9
division:1
Modulus:0
```

5. TO PRINT NAME:

```
class Main
{
public static void main(String[] args)
{
System.out.println("Sivarangjenii");
}
}
```

```
Sivarangjenii

...Program finished with exit code 0
Press ENTER to exit console.
```

6.FIND STUDENTS RESULT USING IF , ELSE AND ELSE IF CONDITION:

```
Main.java
class Main
{
    public static void main(String[] args)
    {
        int m1 = 98;
        int m2 = 97;
        int m3 = 96;
        int m4 = 100;
        int m5 = 100;
        int total = m1+m2+m3+m4+m5;
        int average = total/5;
        System.out.println("total marks: "+total);
        System.out.println("average: "+average);
        if(m1>=35 && m2>=35 && m3>=35 && m4>=35 && m5>=35)
        {
            System.out.println("pass");
        }
        else
        {
            System.out.println("fail");
        }
        if(average>=90)
        {
            System.out.println("Grade: S");
        }
        else if (average>=80)
```


```
            System.out.println("pass");
        }
        else
        {
            System.out.println("fail");
        }
        if(average>=90)
        {
            System.out.println("Grade: S");
        }
        else if (average>=80)
```


```
        {
            System.out.println("Grade: A");
        }
        else
        {
            System.out.println("Grade: B");
        }

    }
}
```

```
total marks: 491
average: 98
pass
Grade: S
```

CLASS PRACTICE SUMS**(14.07.2026)** (PATTERNS USING LOOPS)

1.RIGHT TRIANGLE

PROGRAM EDITOR

```
public class Main
{
  public static void main(String args[])
  {
    int n=4;
    for(int i=1;i<=n;i++)
    {
    for(int j=1;j<=i;j++)
    {
      System.out.print("* ");
    }
      System.out.println();
    }
  }
}
```

OUTPUT

```
*
* *
* * *
* * * *
```

2.LEFT TRIANGLE

```
public class Main
{
  public static void main(String args[])
  {
    int n=4;
    for(int i=1;i<=n;i++)
    {
    for(int j=1;j<=n-i;j++)
    {
      System.out.print("* ");
    }
      System.out.println();
    }
  }
}
```

OUTPUT

```
* * *
* *
*
```

3.PYRAMID

```
public class Main
{
  public static void main(String args[])
  {
    int n=4;
    for(int i=1;i<=n;i++)
    {
    for(int j=1;j<=n-i;j++)
    {
      System.out.print(" ");
    }
      for(int k=1;k<=i;k++)
      {
        System.out.print("* ");
      }
      System.out.println();
    }
  }
}
```

OUTPUT

```
   *
  * *
 * * *
* * * *
```

4.INVERTED PYRAMID

```
public class Main
{
  public static void main(String args[])
  {
    int n=5;
    for(int i=n;i>=1;i--)
    {
    for(int j=1;j<=n-i;j++)
    {
      System.out.print(" ");
    }
      for(int k=1;k<=(2*i-1);k++)
      {
        System.out.print("*");
      }
      System.out.println();
    }
  }
}
```

OUTPUT

```
*********
 *******
  *****
   ***
    *
```

5.DIAMOND

Main.java

```java
import java.util.Scanner;

public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int n = sc.nextInt();

        // Upper half
        for(int i = 1; i <= n; i++)
        {
            for(int j = 1; j <= n - i; j++)
            {
                System.out.print(" ");
            }

            for(int j = 1; j <= 2 * i - 1; j++)
            {
                System.out.print("*");
```

```java
                System.out.print("*");
            }

            System.out.println();
        }

        // Lower half
        for(int i = n - 1; i >= 1; i--)
        {
            for(int j = 1; j <= n - i; j++)
            {
                System.out.print(" ");
            }

            for(int j = 1; j <= 2 * i - 1; j++)
            {
                System.out.print("*");
            }

            System.out.println();
        }
```

```java
        }

        sc.close();
    }
}
```

```
Output

Enter number of rows: 5
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *

=== Code Execution Successful ===
```

6.HOLLOW SQUARE

```java
Main.java                                    Share   Run

import java.util.Scanner;

class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the size of the square: ");
        int n = sc.nextInt();

        for (int i = 1; i <= n; i++)
        {
            for (int j = 1; j <= n; j++)
            {
                if (i == 1 || i == n || j == 1 || j == n)
                    System.out.print("* ");
                else
                    System.out.print("  ");
            }
            System.out.println();
        }
```

```
Output

Enter the size of the square: 4
* * * *
*     *
*     *
* * * *

=== Code Execution Successful ===
```

7.HOLLOW RECTANGLE

Main.java

```java
import java.util.Scanner;
public class Main
{
	public static void main(String[] args)
	{
		Scanner sc = new Scanner(System.in);
		int rows = sc.nextInt();
		int cols = sc.nextInt();

		for(int i=1;i<=rows;i++)
		{
			for(int j=1;j<=cols;j++)
			{
				if(i==1||i==rows||j==1||j==cols)
					System.out.print("*");
				else
					System.out.print(" ");
			}
			System.out.println();
		}

		sc.close();
	}
}
```

```
4 6
******
*    *
*    *
******
```

8.SQUARE

Main.java

```java
import java.util.Scanner;
public class Main
{
	public static void main(String[] args)
	{
		Scanner sc=new Scanner(System.in);
		int n=sc.nextInt();

		for(int i=1;i<=n;i++)
		{
			for(int j=1;j<=n;j++)
			{
				System.out.print("*");
			}
			System.out.println();
		}

		sc.close();
	}
}
```

```
4
****
****
****
****
```

9.RECTANGLE

```java
import java.util.Scanner;
public class Main
{
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        int rows=sc.nextInt();
        int cols=sc.nextInt();

        for(int i=1;i<=rows;i++)
        {
            for(int j=1;j<=cols;j++)
            {
                System.out.print("*");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

```
4 8
********
********
********
********
```

10.NUMBER PATTERN:

Write a program to print the below pattern?

1

1 1

1 2 1

1 3 3 1

1 4 6 4 1

Main.java

```java
import java.util.Scanner;

public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        for(int i = 0; i < n; i++)
        {
            int num = 1;

            for(int j = 0; j <= i; j++)
            {
                System.out.print(num + " ");
                num = num * (i - j) / (j + 1);
            }

            System.out.println();
        }

        sc.close();
    }
}
```

```
5
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

11. Write a program to print the following pattern

Sample Input:

Enter the number to be printed: 1

Max Number of time printed: 3

1

11

111

11

1

```java
import java.util.Scanner;

public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number to be printed: ");
        int num = sc.nextInt();

        System.out.print("Max Number of time printed: ");
        int n = sc.nextInt();

        for(int i = 1; i <= n; i++)
        {
            for(int j = 1; j <= i; j++)
            {
                System.out.print(num);
            }
            System.out.println();
        }

        for(int i = n - 1; i >= 1; i--)
        {
            for(int j = 1; j <= i; j++)
            {
                System.out.print(num);
            }
            System.out.println();
        }

        sc.close();
    }
}
```

```
Enter the number to be printed: 1
Max Number of time printed: 3
1
11
111
11
1
```

12. Write a program to print the following pattern

Sample Input:

Enter the Character to be printed: %

Max Number of time printed: 3

%

% %

% % %

Main.java

```java
import java.util.Scanner;

public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the Character to be printed: ");
        char ch = sc.next().charAt(0);

        System.out.print("Max Number of time printed: ");
        int n = sc.nextInt();

        for(int i = 1; i <= n; i++)
        {
            for(int j = 1; j <= i; j++)
            {
                System.out.print(ch + " ");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

input

```
Enter the Character to be printed: %
Max Number of time printed: 3
%
% %
% % %
```

13. Write a program to print the below pattern

1

2 2

3 3 3

4 4 4 4

Main.java

```java
import java.util.Scanner;

public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        for(int i=1;i<=n;i++)
        {
            for(int j=1;j<=i;j++)
            {
                System.out.print(i+" ");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

```
4
1
2 2
3 3 3
4 4 4 4
```

14. Write a program to print the below pattern

1

4 9

16 25 36

49 64 81 100

Main.java

```java
import java.util.Scanner;

public class Main
{
	public static void main(String[] args)
	{
		Scanner sc = new Scanner(System.in);
		int n = sc.nextInt();

		int num = 1;

		for(int i=1;i<=n;i++)
		{
			for(int j=1;j<=i;j++)
			{
				System.out.print(num*num+"  ");
				num++;
			}
			System.out.println();
		}

		sc.close();
	}
}
```

```
4
1
4  9
16  25  36
49  64  81  100
```

15. Write a program to print the below pattern

1

2 2

3 3 3

4 4 4 4

3 3 3

2 2

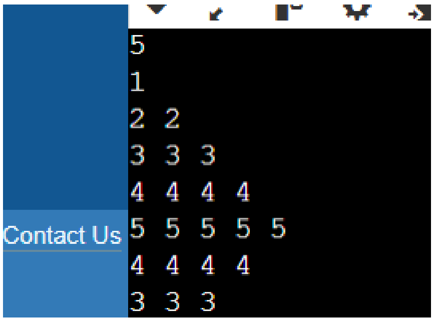
1

Main.java

```java
import java.util.Scanner;

public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        for(int i=1;i<=n;i++)
        {
            for(int j=1;j<=i;j++)
            {
                System.out.print(i+" ");
            }
            System.out.println();
        }

        for(int i=n-1;i>=1;i--)
        {
            for(int j=1;j<=i;j++)
            {
                System.out.print(i+" ");
            }
            System.out.println();
        }
```


```java
            {
                System.out.print(i+" ");
            }
            System.out.println();
        }
```


```java

        sc.close();
    }
}
```



16.SUM OF FIBONACCI SERIES

Main.java

```java
import java.util.Scanner;

public class Main {

    static int fibonacciSum(int n) {
        int a = 0, b = 1, c;
        int sum = 0;

        for (int i = 1; i <= n; i++) {
            sum = sum + a;
            c = a + b;
            a = b;
            b = c;
        }

        return sum;
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        System.out.println(fibonacciSum(n));

        sc.close();
    }
```

```
5
7


...Program finished with exit code 0
Press ENTER to exit console.
```

17.